

ZERO DEGREE CALORIMETER

[G4 User Guide](#)

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Cheat sheet: <http://cmsdoc.cern.ch/~belt/CheatSheet.html>

Most Recent Working Directory

data/users/jbwells4/jbwells4/ZDCTDR_NEW_Cerenkov/

Initial Setup Scripts

(run EVERY time)

```
source /data/users/belt/geant4/setup/setup.csh
source /data/users/belt/geant4/setup/setup_geant_opengl.csh
setenv G4NEUTRONHPDATA /data/users/jgomez2/g4course2010/BROND-2.2
```

Run Command

(Visualizer Module)

```
$G4WORKDIR/bin/Linux-g++/HitDevel
```

How to write text output from a GEANT command to a text file instead of the screen

```
$G4WORKDIR/bin/Linux-g++/HitDevel runPgun_test.mac >& test_file_output2.txt
```

This is using a UNIX trick, where **>&** put the stdout and stderr into a file & puts the whole process running in background.

It may not continue running if you "exit" that UNIX login session (or your laptop runs out of network connection).

Working Scripts

runPgun_10_batch.mac	//contains centered RPD gun positions
runPgun_10_batch_RPDTest.mac	//contains 4 positions per RPD, 1st quadrant

Run Command Example

```
$G4WORKDIR/bin/Linux-g++/HitDevel runPgun_10_batch.mac > &  
test_file_RPDTEST_n_10_Aug_31.txt
```

Basic Run Protocol

(As I gather)

- DetectorConstruction class creates volumes and solid objects
- HistoManager class initializes histograms
- RunAction class loops over all events
 - EventAction class loops over all Steps
 - SteppingAction class creates Tracks, made up of Steps
 - PrimaryGenerator and PhysicsList initialize simulation of primary particles
 - CaloHit/CalorimeterHit and SimpleHit initialize simulation of secondary and granddaughter++ instances with PhysicsList reference
 - Fills "Shower Profile" with all relevant Tracks
 - RunAction fills all other histos with energy accumulated in RPD per Event
- Methods called by RunAction contained in EventAction and HistoManager (see #include lists)

******My best friend in this project was `>grep -r` and `splat!` Just find the relevant method calls, then search for them in the `src/` and `include/` directories. Good Luck.....

ROOT commands. / Hist

We use the CMSSW (CMS software) working environment which has root built in.

1) Create a new CMSSW working area (this is the version I am using right now for some CMS software stuff) - this needs to be done ONLY ONCE, as it creates a directory structure:
`cmsrel CMSSW_5_3_16`

2) This needs to be done every time you login and want to run root *For that login session*:

```
cd CMSSW_5_3_16/src
cmsenvroot
```

Please note as I told you that you don't want to have a "cmsenv" in the same login window in which you have setup the geant4 environment. I typically have two terminal windows open, one for geant4, one for cmssw/root.

ROOT documentation:

<http://root.cern.ch>